# Python Unit Testing Activity Report

Course: SWEN-352 *Software Testing*
Team: Team 3
Professor: Bryan Basham

## Select a Component

- Devin Rhodie - Library
- Yousaf Iqbal – DB Interface
- Haku Iijima– API Interface

## Source Repo

GitHub URL: [https://github.com/RIT-SWEN-352/python-unit-testing-team-3-1](https://github.com/RIT-SWEN-352/python-unit-testing-team-3-1)

# Steps #1 & 2: Configure Python Virtual Environment

Haku Iijima:

```
hakui@TABLET-2TGM354B MINGW64 ~/swen352/python-unit-testing-team-3-1 (main)
$ which python
(.venv)
hakui@TABLET-2TGM354B MINGW64 ~/swen352/python-unit-testing-team-3-1 (main)
$ pip install -r requirements.txt
Collecting atomicwrites==1.3.0
  Downloading atomicwrites-1.3.0-py2.py3-none-any.whl (5.9 kB)
Collecting attrs==18.2.0
  Downloading attrs-18.2.0-py2.py3-none-any.whl (34 kB)
Collecting certifi==2024.6.2
  Downloading certifi-2024.6.2-py3-none-any.whl (164 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 164.4/164.4 KB 3.3 MB/s eta 0:00:00
Collecting chardet==3.0.4
  Downloading chardet-3.0.4-py2.py3-none-any.whl (133 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 133.4/133.4 KB 7.7 MB/s eta 0:00:00
Collecting coverage==7.5.3
  Downloading coverage-7.5.3-cp39-cp39-win_amd64.whl (207 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 207.4/207.4 KB 12.3 MB/s eta 0:00:00
Collecting idna==3.7
  Downloading idna-3.7-py3-none-any.whl (66 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 66.8/66.8 KB ? eta 0:00:00
Collecting more-itertools==6.0.0
  Downloading more_itertools-6.0.0-py3-none-any.whl (52 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 52.4/52.4 KB 2.6 MB/s eta 0:00:00
Collecting pbr==5.1.2
  Downloading pbr-5.1.2-py2.py3-none-any.whl (107 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 107.1/107.1 KB ? eta 0:00:00
```

```
hakui@TABLET-2TGM354B MINGW64 ~/swen352/python-unit-testing-team-3-1 (main)
$ pip freeze
atomicwrites==1.3.0
attrs==18.2.0
certifi==2024.6.2
chardet==3.0.4
charset-normalizer==3.4.4
coverage==7.5.3
idna==3.7
more-itertools==6.0.0
pbr==5.1.2
pluggy==0.8.1
py==1.8.0
requests==2.32.3
six==1.12.0
tinydb==4.8.0
urllib3==2.2.1
(.venv)
```

Yousaf Iqbal:

```
(.venv)
yousa@Yousaf CLANGARM64 ~/SE/Spring 2026/python-unit-testing-team-3-1 (main)
$ python --version
Python 3.14.2
(.venv)
yousa@Yousaf CLANGARM64 ~/SE/Spring 2026/python-unit-testing-team-3-1 (main)
$ pip freeze
atomicwrites==1.3.0
attrs==18.2.0
certifi==2024.6.2
chardet==3.0.4
charset-normalizer==3.4.4
coverage==7.5.3
idna==3.7
more-itertools==6.0.0
pbr==5.1.2
pluggy==0.8.1
py==1.8.0
requests==2.32.3
six==1.12.0
tinydb==4.8.0
urllib3==2.2.1
(.venv)
yousa@Yousaf CLANGARM64 ~/SE/Spring 2026/python-unit-testing-team-3-1 (main)
$
```

Devin Rhodie:

SWEN352-Activity-2

PROBLEMS OUTPUT TERMINAL bash

```
(.venv)
15164@Devins-Laptop MINGW64 ~/github-classroom/python-unit-testing-team-3-1 (main)
$ pip install -r requirements.txt
(.venv)
15164@Devins-Laptop MINGW64 ~/github-classroom/python-unit-testing-team-3-1 (main)
$ python --version
Python 3.9.13
(.venv)
15164@Devins-Laptop MINGW64 ~/github-classroom/python-unit-testing-team-3-1 (main)
$ pip freeze
atomicwrites==1.3.0
attrs==18.2.0
certifi==2024.6.2
chardet==3.0.4
charset-normalizer==3.4.4
coverage==7.5.3
idna==3.7
more-itertools==6.0.0
pbr==5.1.2
pluggy==0.8.1
py==1.8.0
requests==2.32.3
six==1.12.0
tinydb==4.8.0
urllib3==2.2.1
(.venv)
15164@Devins-Laptop MINGW64 ~/github-classroom/python-unit-testing-team-3-1 (main)
$
```

# Steps #3, 4, 5, & 6: Individual Coverage Report

Yousaf Iqbal:

Coverage report: 98%

Files | Functions | Classes

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

| File ▲ | statements | missing | excluded | coverage |
|---|---|---|---|---|
| library\library_db_interface.py | 40 | 1 | 0 | 98% |
| **Total** | **40** | **1** | **0** | **98%** |

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

Coverage report: 98%

Files | Functions | Classes

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

| File ▲ | function | statements | missing | excluded | coverage |
|---|---|---|---|---|---|
| library\library_db_interface.py | Library_DB.__init__ | 3 | 1 | 0 | 67% |
| library\library_db_interface.py | Library_DB.insert_patron | 6 | 0 | 0 | 100% |
| library\library_db_interface.py | Library_DB.get_patron_count | 2 | 0 | 0 | 100% |
| library\library_db_interface.py | Library_DB.get_all_patrons | 2 | 0 | 0 | 100% |
| library\library_db_interface.py | Library_DB.update_patron | 6 | 0 | 0 | 100% |
| library\library_db_interface.py | Library_DB.retrieve_patron | 5 | 0 | 0 | 100% |
| library\library_db_interface.py | Library_DB.close_db | 1 | 0 | 0 | 100% |
| library\library_db_interface.py | Library_DB.convert_patron_to_db_format | 1 | 0 | 0 | 100% |
| library\library_db_interface.py | *(no function)* | 14 | 0 | 0 | 100% |
| **Total** | | **40** | **1** | **0** | **98%** |

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

Devin Rhodie:

## Coverage report: 98%

Files | Functions | Classes

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

| File ▲ | statements | missing | excluded | coverage |
|---|---|---|---|---|
| library\library.py | 53 | 0 | 0 | 100% |
| **Total** | **53** | **0** | **0** | **100%** |

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

## Coverage report: 98%

Files | Functions | Classes

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

| File ▲ | function | statements | missing | excluded | coverage |
|---|---|---|---|---|---|
| library\library.py | Library.__init__ | 2 | 0 | 0 | 100% |
| library\library.py | Library.shutdown | 1 | 0 | 0 | 100% |
| library\library.py | Library.is_ebook | 6 | 0 | 0 | 100% |
| library\library.py | Library.get_ebooks_count | 5 | 0 | 0 | 100% |
| library\library.py | Library.is_book_by_author | 6 | 0 | 0 | 100% |
| library\library.py | Library.get_languages_for_book | 6 | 0 | 0 | 100% |
| library\library.py | Library.register_patron | 2 | 0 | 0 | 100% |
| library\library.py | Library.is_patron_registered | 4 | 0 | 0 | 100% |
| library\library.py | Library.borrow_book | 2 | 0 | 0 | 100% |
| library\library.py | Library.return_borrowed_book | 2 | 0 | 0 | 100% |
| library\library.py | Library.is_book_borrowed | 2 | 0 | 0 | 100% |
| library\library.py | *(no function)* | 15 | 0 | 0 | 100% |
| **Total** | | **53** | **0** | **0** | **100%** |

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

Haku Iijima:

## Coverage report: 98%

Files | Functions | Classes

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

| File ▲ | statements | missing | excluded | coverage |
|---|---|---|---|---|
| library\ext_api_interface.py | 47 | 2 | 0 | 96% |
| **Total** | **47** | **2** | **0** | **96%** |

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

## Coverage report: 98%

Files | Functions | Classes

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*

| File ▲ | function | statements | missing | excluded | coverage |
|---|---|---|---|---|---|
| library\ext_api_interface.py | Books_API.is_book_available | 4 | 0 | 0 | 100% |
| library\ext_api_interface.py | Books_API.books_by_author | 7 | 0 | 0 | 100% |
| library\ext_api_interface.py | Books_API.get_book_info | 12 | 0 | 0 | 100% |
| library\ext_api_interface.py | Books_API.get_ebooks | 8 | 1 | 0 | 88% |
| library\ext_api_interface.py | Books_API.gather_test_data | 1 | 1 | 0 | 0% |
| library\ext_api_interface.py | Books_API.__make_request | 6 | 0 | 0 | 100% |
| library\ext_api_interface.py | *(no function)* | 9 | 0 | 0 | 100% |
| **Total** | | **47** | **2** | **0** | **96%** |

*coverage.py v7.5.3, created at 2026-02-12 14:58 -0500*